



IoT is Smart for Buildings

By Christopher Weems

9

Buildings are increasingly no longer just containers for life. Through technology, buildings are gaining lives of their own. With the amount of data that can be accessed about everything from occupancy, to airflow rates, to energy use, it is almost as if buildings are nearing sentience. And, as in science-fiction accounts of inanimate things achieving sentience, it spreads rapidly.

“Buildings are becoming data-rich environments as owners and key decision makers continue to invest in a variety of solutions that embody the technology foundation of the Internet of Things (IoT) and cloud computing,” says industry research provider Navigant Research in a recently released report. The [“Intelligent Buildings: 10 Trends to Watch in 2016 and Beyond” report](#) goes on to state “As building energy management systems (BEMSs) gain greater market penetration, building control algorithms are beginning to provide data beyond energy efficiency, including details on space utilization and operational efficiency.”



Intelligent Buildings: 10 Trends to Watch in 2016 and Beyond

The “Intelligent Buildings: 10 Trends to Watch in 2016” report by Navigant Research.

London-based Memoori Business Intelligence defines a subset of IoT that it calls the Building Internet of Things (BloT) as: “The overlaying of an IP network, connecting all the building services monitoring, analyzing and controlling without the intervention of humans.”

Johnson Controls (JCI) is placing a lot of stock in smart buildings, which it calls “agile buildings.” The company uses the word “agile” to describe a building that is “able to *gracefully* adapt to change.” During the 2016 AHR Expo, held earlier this year, JCI held a session to discuss smart building and IoT topics.



Johnson Controls' "2016 Energy Efficiency Indicator Survey."

During the discussion, Nazee Hoglund, vice president – systems product management, Johnson Controls, said “As building systems move forward, it’s increasingly important that they communicate with other systems, including mobile phones and other devices. This way we can change the dynamics of our business by providing useful information that improves the business outcome.”

In its recently released “[2016 Energy Efficiency Indicator Survey](#),” JCI concluded that businesses involved in the built environment are indeed taking a good look at how making their building stock “smarter” is affecting their current and future business outcomes. Seventy-two percent of “large organizations” (with building stock of more than 100,000 ft²) respondents to the survey said they had installed a building management system (BMS).

Not Moving at the Speed of the Internet

But, are current buildings (and more importantly, their owners) truly agile? Are buildings keeping up with adopting the latest technologies? Memoori’s data seems to suggest that the in the race to adopt IoT technologies, the buildings industry is more tortoise than hare. Its “[The Internet of Things in Smart Buildings 2014 to 2020](#)” report concluded that

mainstream penetration, i.e. 50% of all building systems devices connected, is unlikely to be achieved before 2025.

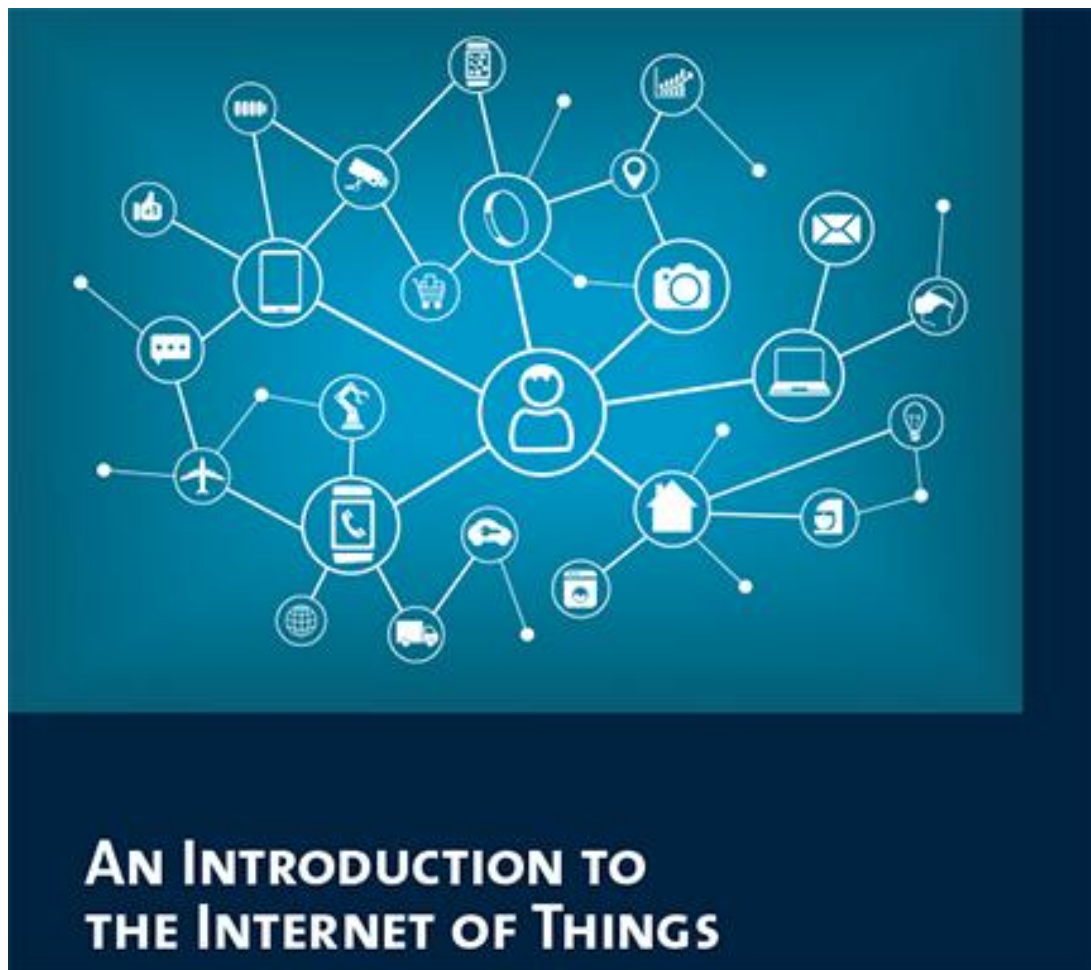
There's information technology speed, and buildings industry speed. Making wholesale changes to existing buildings, by their nature, is difficult. And, around 75% of the projected building stock of 2050 exists today. Also, businesses must consider the cost impacts. The average cost to deploy a basic BMS is at least \$2.50 per square foot and can be as high as \$7 per square foot, according to the Cooperative Research Network of the National Rural Electric Cooperative Association (NRECA). This equates to at least \$250,000 for a 100,000 ft² building.

The Business Case for Smart Buildings

Looking at the financial benefits, rather than the costs, can likely spur more building owners and facility managers to adopt and implement IoT technologies in their building portfolios. Digital and social strategist Gunjan Rawal says that it is reasonable to expect savings in the range of 10% to 25% when implementing proactive energy-management programs in mid-sized buildings. She adds that return on investment (ROI) from properly-applied IoT sensors, switches, and analytics can occur in six months to two years, a fraction of the time it takes to recoup investment in traditional BMS systems.

Another key to the adoption of IoT technologies will be its development as a necessity, rather than a novelty. The buildings industry may not be ahead of the curve in comparison to the rest of society, but it must remain close to the larger societal paradigm.

According to Underwriters Laboratories (UL), "future growth of the IoT will come from applications of the IoT technology in virtually every sector of the economy, from commercial and industrial environments to health care and public safety."



UL's "An Introduction to the Internet of Things" white paper.

A close relative to the buildings industry is manufacturing. Manufacturing is also moving towards becoming smarter.

According to Cory Vanderpool, Business Development Director at Magnum Energy, **"The potential for manufacturers to collect key data from various physical spaces within the plant is enormous."** She says that some of the key uses of IoT in manufacturing settings include sensors for alerting to problems on the production line, tracking of assets in the production process, and extracting data about the time and quantity of items dispensed by machines in the plant.

"The IoT has the potential to dramatically increase the availability of information, and is likely to transform companies and organizations in virtually every industry around the world," writes UL in its **"An Introduction to the Internet of Things" white paper**. "As such, finding ways to leverage the power of the IoT is expected to factor into the strategic objectives of most technology companies, regardless of their industry focus."

The Internet of Things is here. While it may not be in every building yet, in time, it will be.



High Efficiency

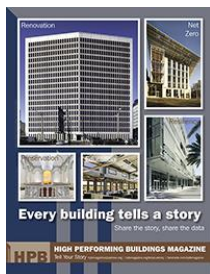
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